Comparison between types of cellular organisation

Features	Prokaryotes	Mesokaryotes	Eukaryotes
Size of the cell	~1-5µm	~5-10µm	~10-100µm
Nuclear character	Nucleoid, no true nucleus,	Nucleus with nuclear membrane	True nucleus with nuclear membrane
DNA	Usually circular without histone proteins	Usually linear but without histone proteins	Usually linear with histone proteins
RNA/Protein synthesis	Couples in cytoplasm	Similar with eukaryotes	RNA synthesis inside nucleus/ Protein synthesis in cytoplasm
Ribosomes	50S+ 30S	60S + 40S	60S + 40S
Organelles	Absent	Present	Numerous
Cell movement	Flagella	Gliding and flagella	Flagella and cilia
Organization	Usually single cell	Single and colony	Single, colonial and multicellular
Cell division	Binary fission	Binary fission	Mitosis and meiosis
Examples	Bacteria and Archaea	Dinoflagellate, Protozoa	Fungi, plants and animals

6.5. Plant and Animal cell

6.5.1 Ultra Structure of an Eukaryotic Cell

An eukaryotic cell is highly distinct in its organisation. It shows several variations in different organisms. For instance, eukaryotic cells in plants and animals vary greatly (Figure 6.7).

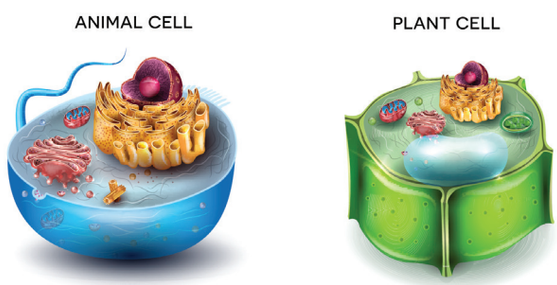


Figure 6.7: Animal and Plant cell

Animal Cell

Animal cells are surrounded by cell membrane or plasma membrane. Inside this membrane a gelatinous matrix called **protoplasm** is seen to contain nucleus and other organelles which include the endoplasmic reticulum, mitochondria, golgi bodies, centrioles, lysosomes, ribosomes and cytoskeleton.

Plant cell

A typical plant cell has prominent cell wall, a large central vacuole and plastids in addition to other organelles present in animal cell (Figure 6.8).

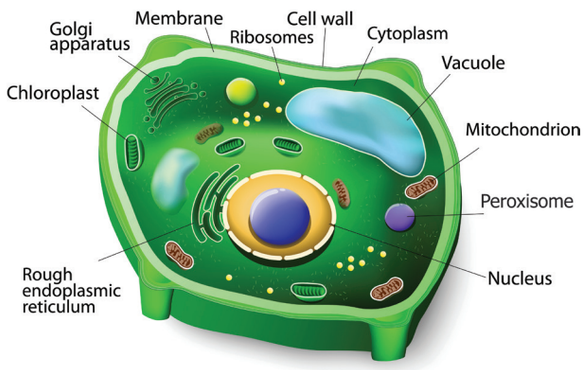


Figure 6.8: Ultra Structure of Plant Cell

6.5.2 Protoplasm

Protoplasm is the living content of cell that is surrounded by plasma membrane. It is a colourless material that exists throughout the cell together with cytoplasm, nucleus and other organelles. Protoplasm is composed of a mixture of small particles, such as

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